

Lab 3: IMU Noise Characterization with Allan Variance

EECE 5554 Robotics Sensing and Navigation

Lab Objective:

To analyze the data collected by VectorNAV-100 IMU sensor and understand the noise characteristics and error distribution in the sensor. Also to make out the sources of error affecting the sensor. Based upon the conclusions drawn answer the following questions :

1. What kind of errors/sources of noise are present?
2. How do we model them? Where do we measure them? Can you relate your measurements to the datasheet for the VN100?

Hardware and Setup

IMU Device

- **Model:** VectorNav VN-100
- **Configuration:**
 - Open a serial port with 115200 baudrate.
 - Configure the IMU to send data at 40 Hz.
 - Ensure data is published in correct units for message fields.

IMU Driver

- Custom Message Package
- Topic to Publish: '/imu'

Methods to Collect Data:

Individually, collecting 10-15 minutes of data using the personal IMU Driver. This data is utilized to plot the time-series and frequency distribution graphs for all 3 axis (x-axis, y-axis, z-axis) of linear acceleration, angular acceleration and magnetic field as well as Roll, Pitch, Yaw for the orientation.

As a group, one person collects data for 5 hours. This data is utilized for plotting Allan-Variations and used to understand the noise characteristics of the IMU sensor.

The link to the 5 Hour data:

<https://drive.google.com/drive/folders/1mWaPnAXSMHJcyhvr0F9ZxtxVoMLDuM7Y>

Data Analysis:

Stationary Noise Analysis

- **Time Series Plots:**
 - **Linear Acceleration (X, Y, Z)**

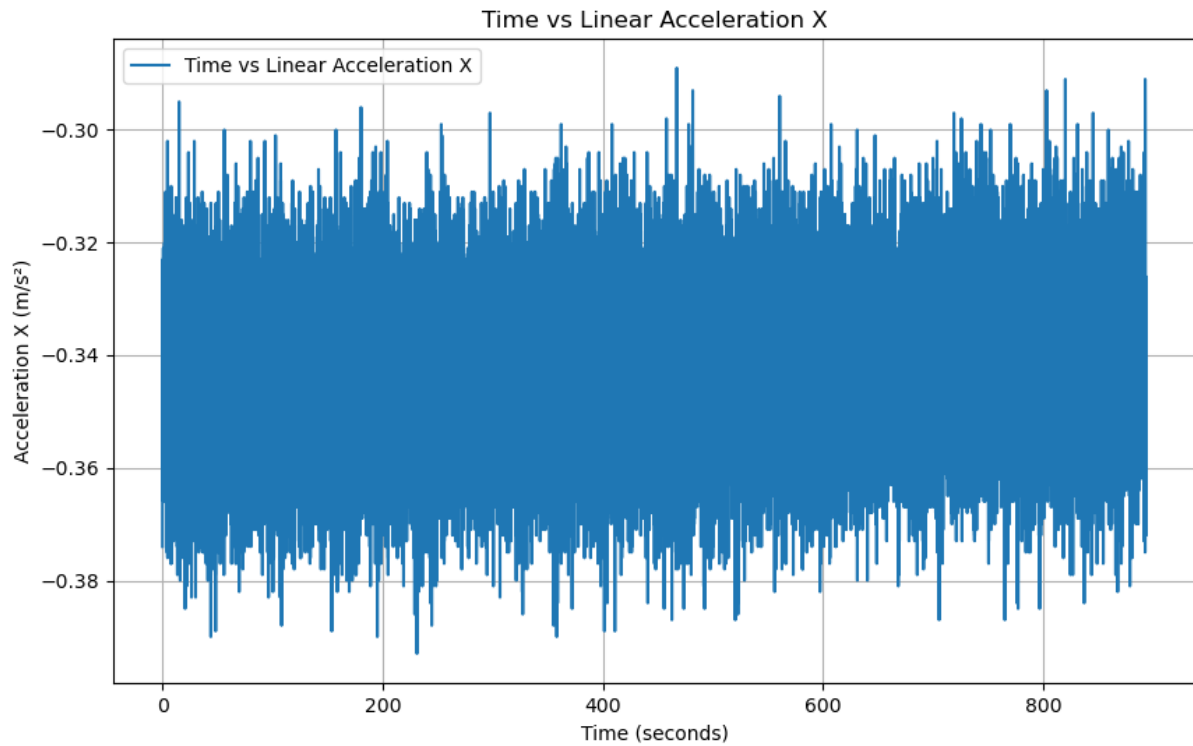


Figure 1: Time series data for accelerometer X-axis.

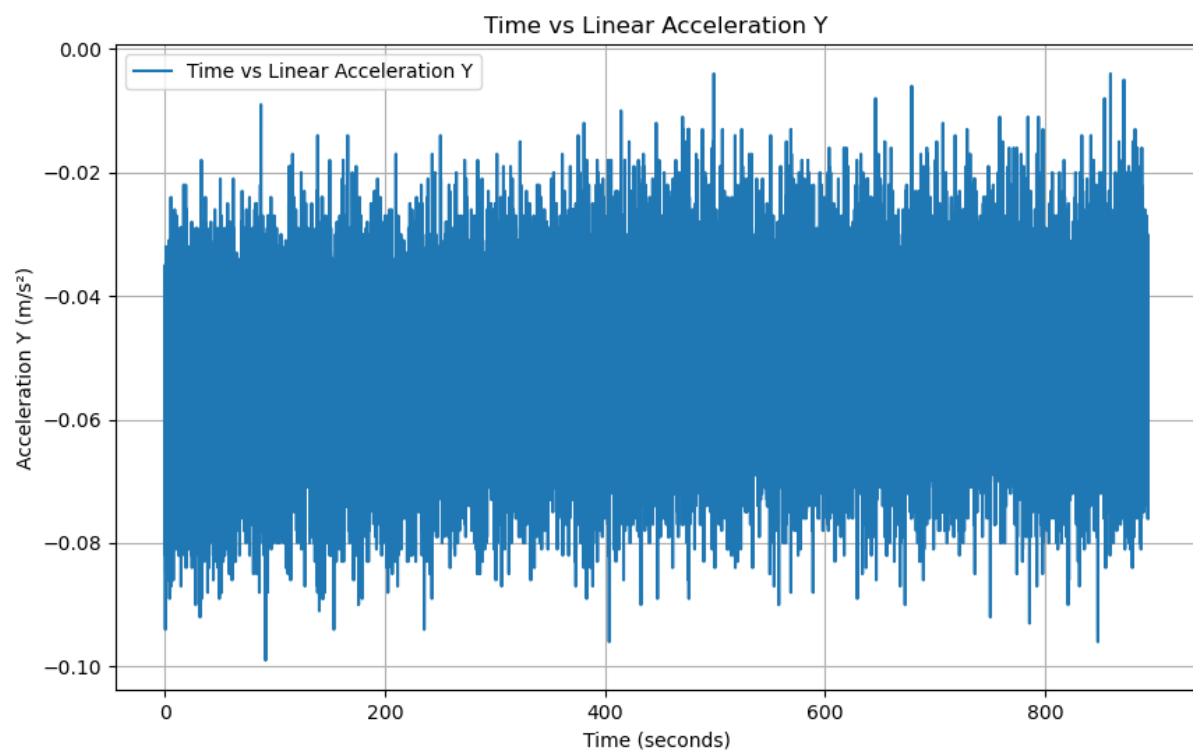


Figure 2: Time series data for accelerometer Y-axis.

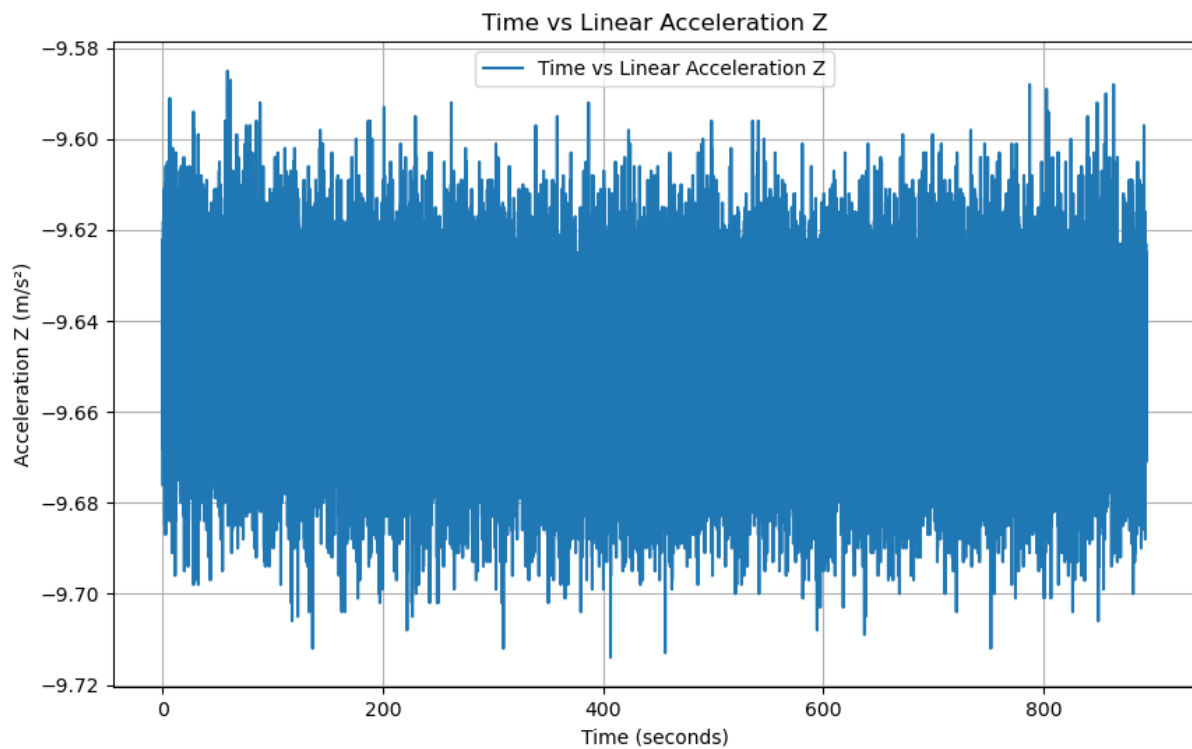


Figure 3: Time series data for accelerometer Z-axis.

○ **Angular Velocity (X, Y, Z)**

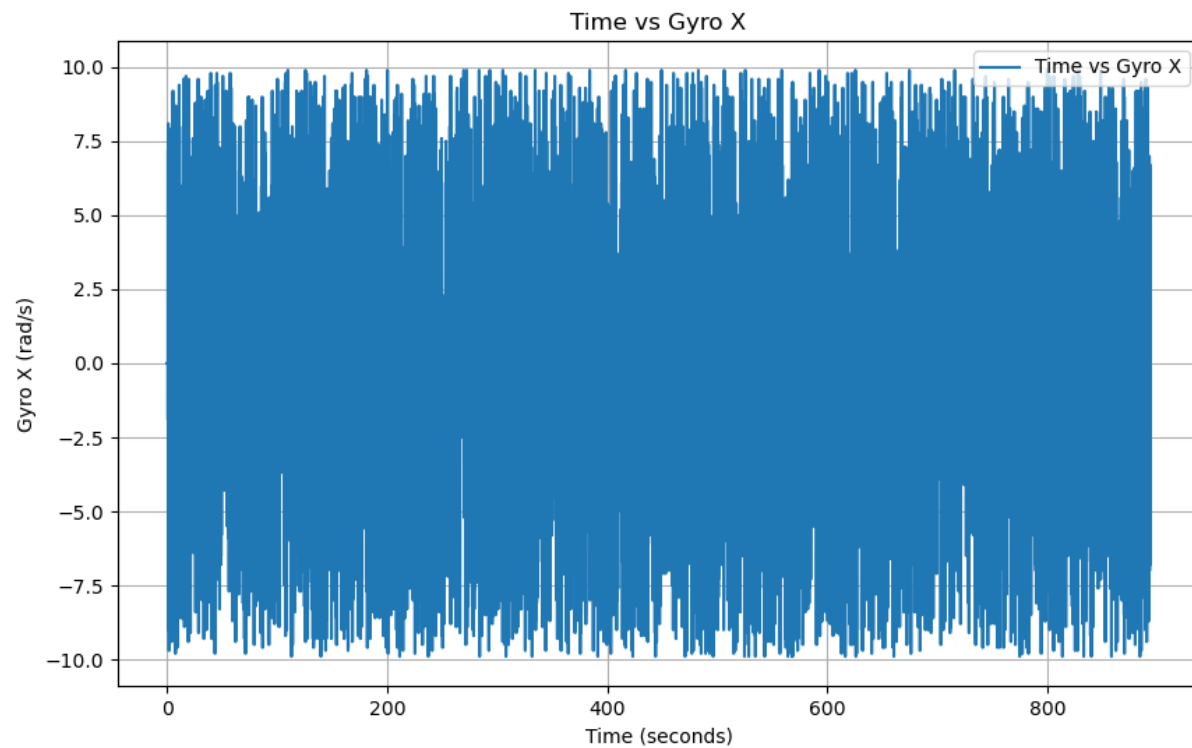


Figure 4: Time series data for gyroscope X-axis.

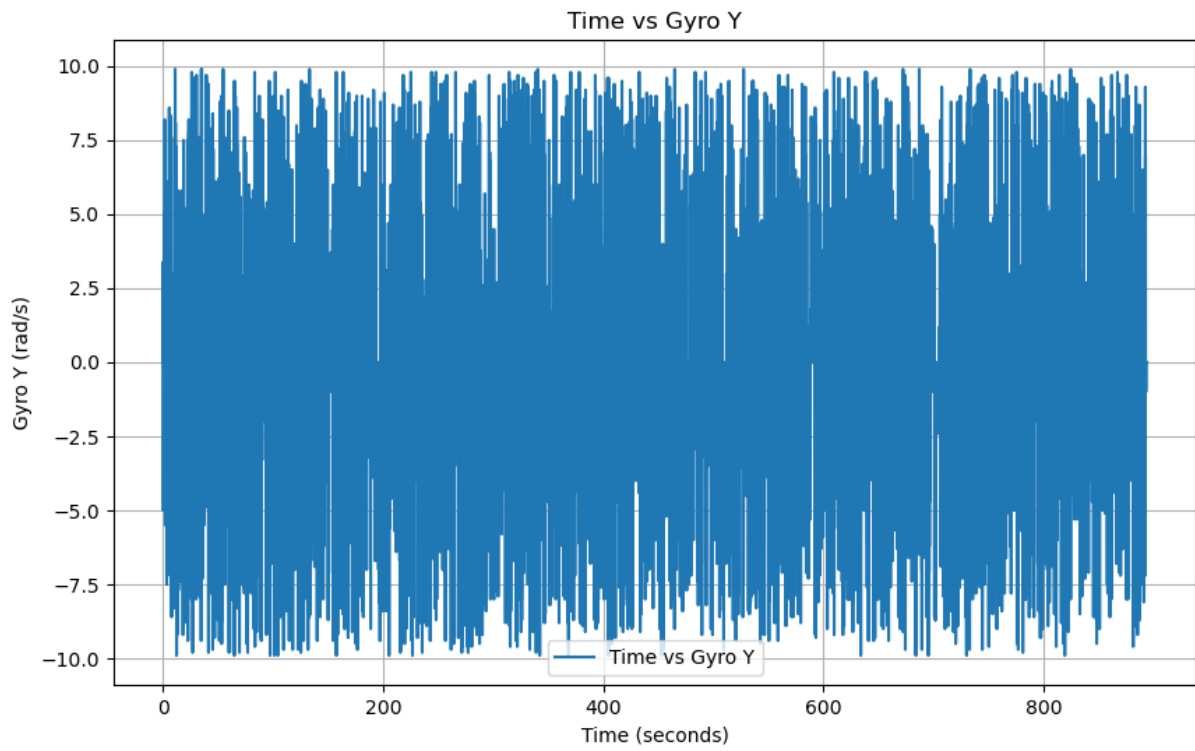


Figure 5: Time series data for gyroscope Y-axis.

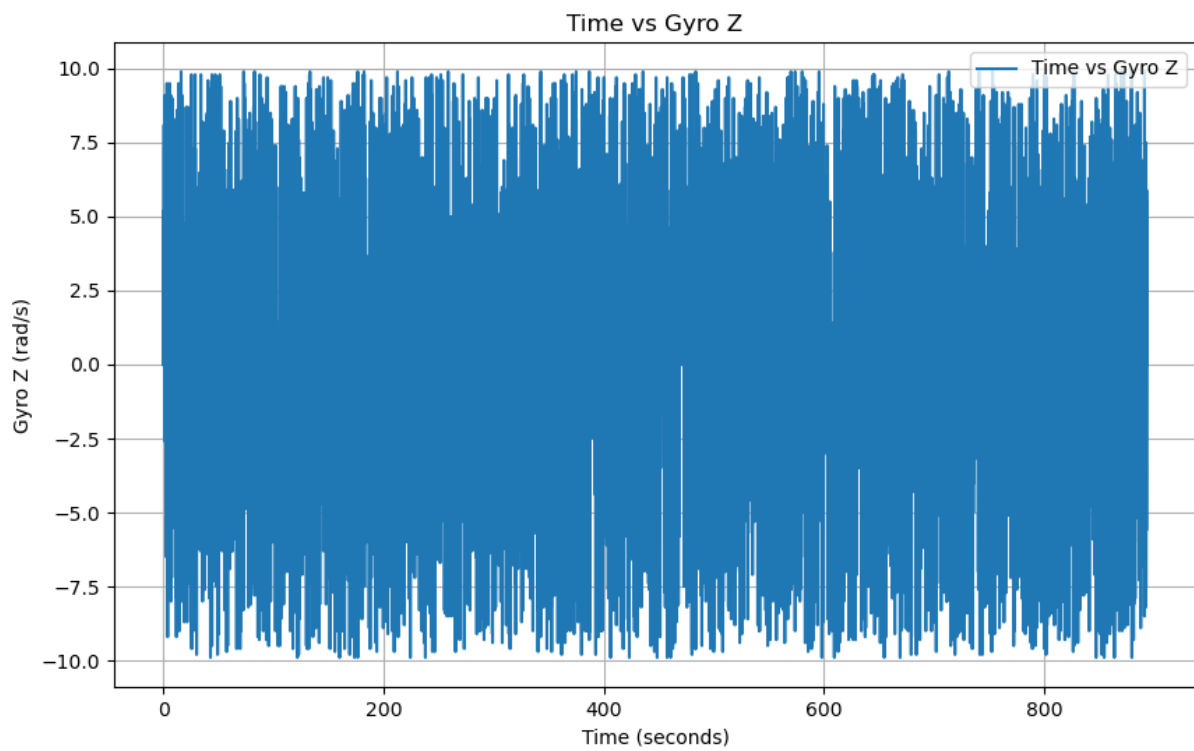


Figure 6: Time series data for gyroscope Z-axis.

- **Magnetic Field (X, Y, Z)**

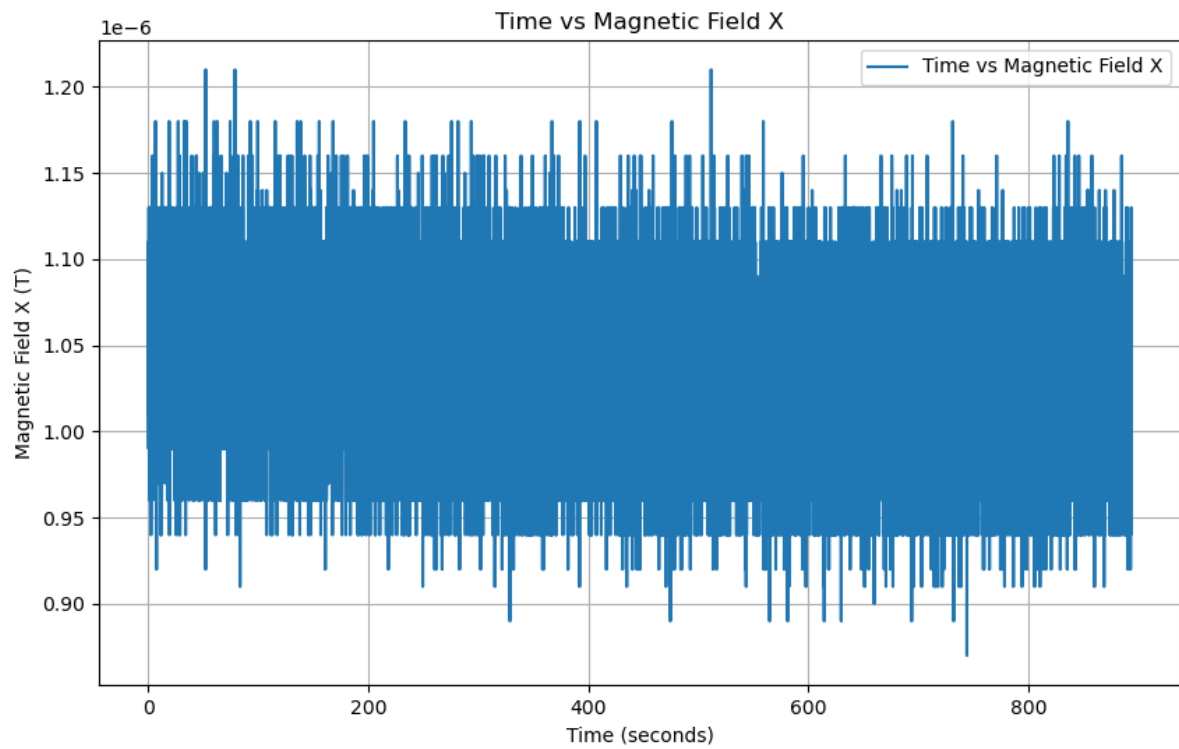


Figure 7: Time series data for magnetometer X-axis.

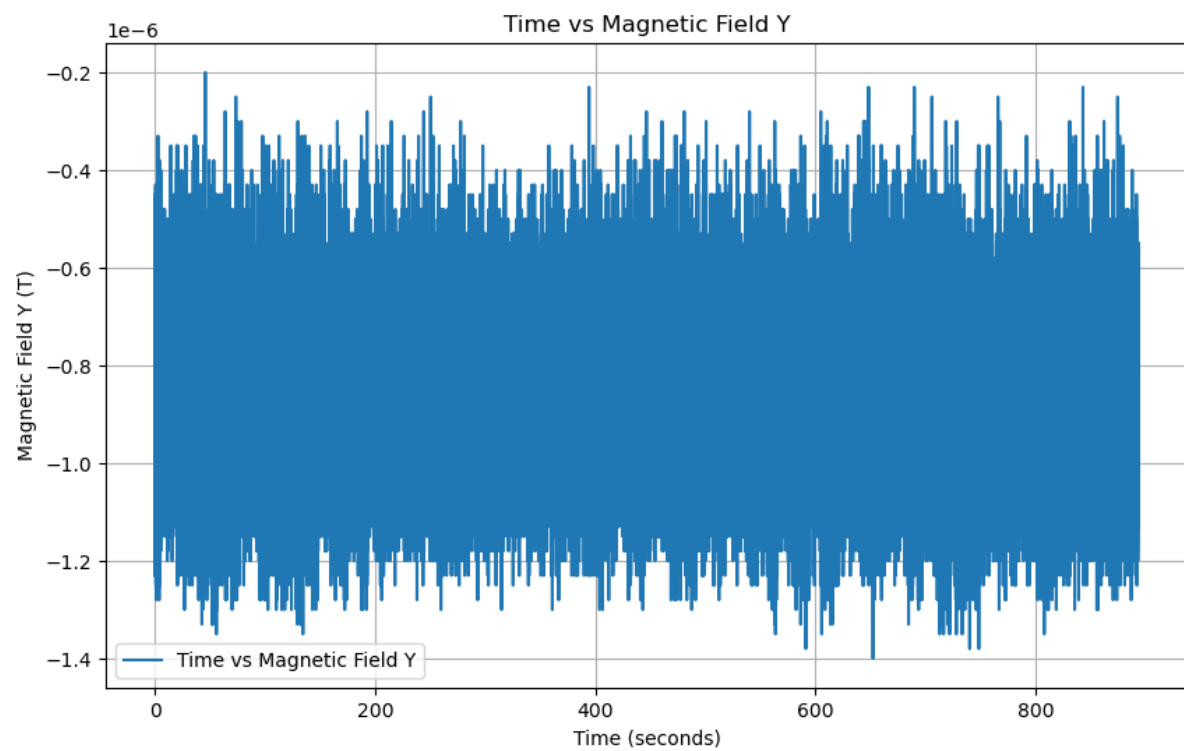


Figure 8: Time series data for magnetometer Y-axis.

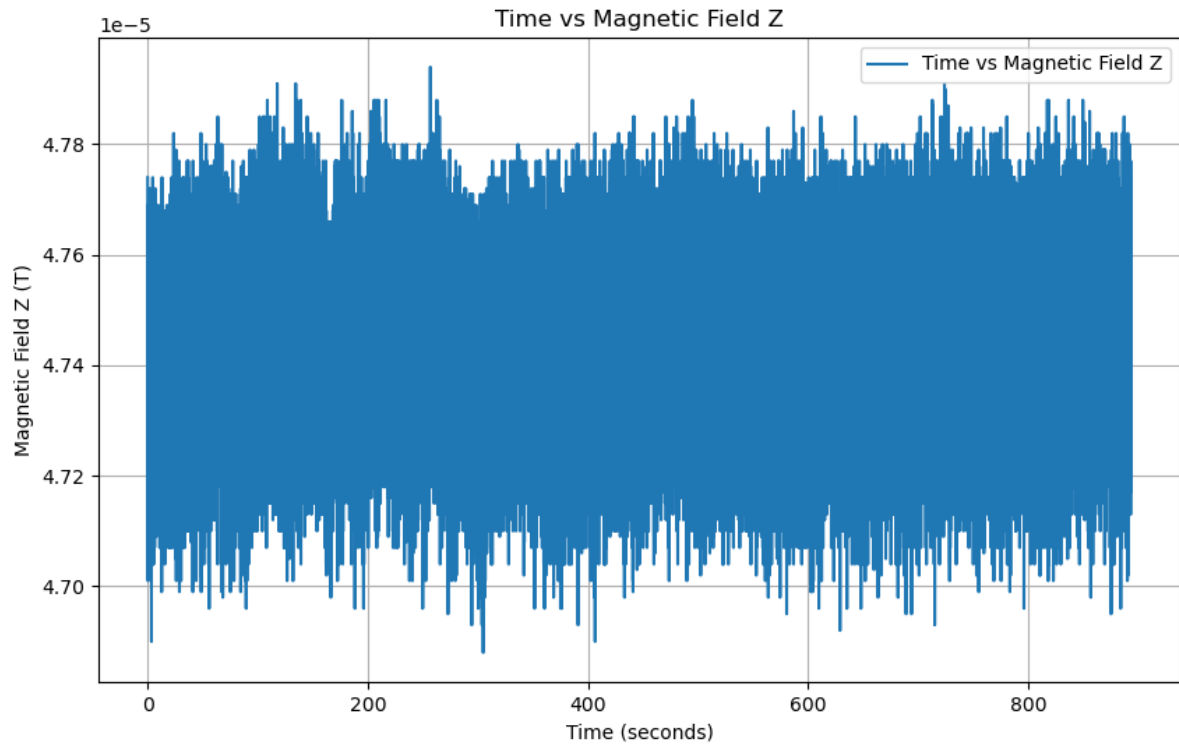


Figure 9: Time series data for magnetometer Z-axis.

- **Orientation (Roll, Pitch, Yaw)**

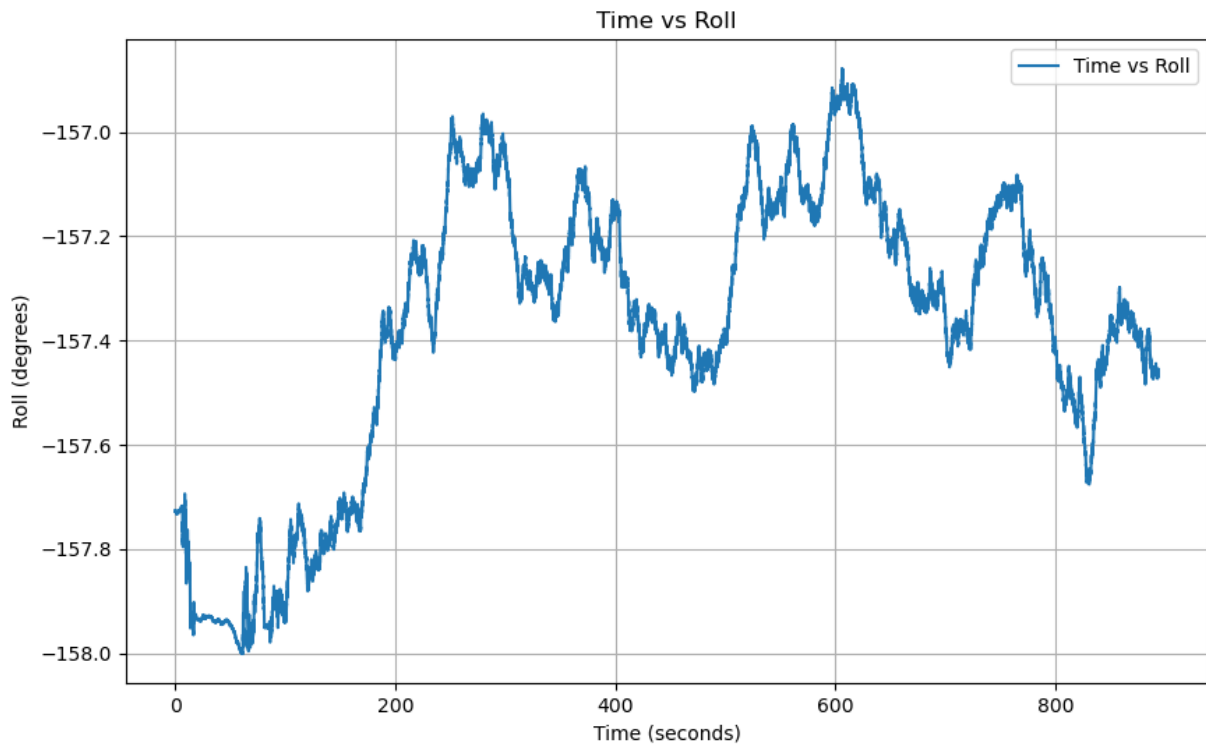


Figure 10: Time series data for Roll.

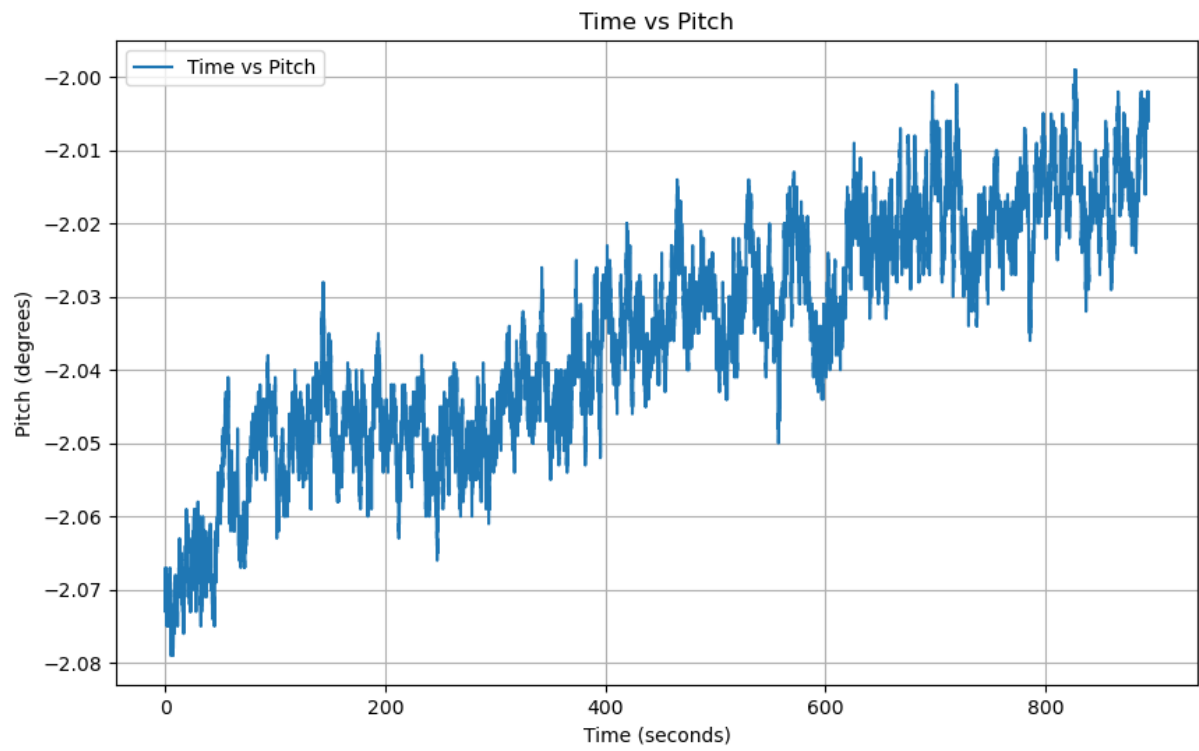


Figure 11: Time series data for Pitch.

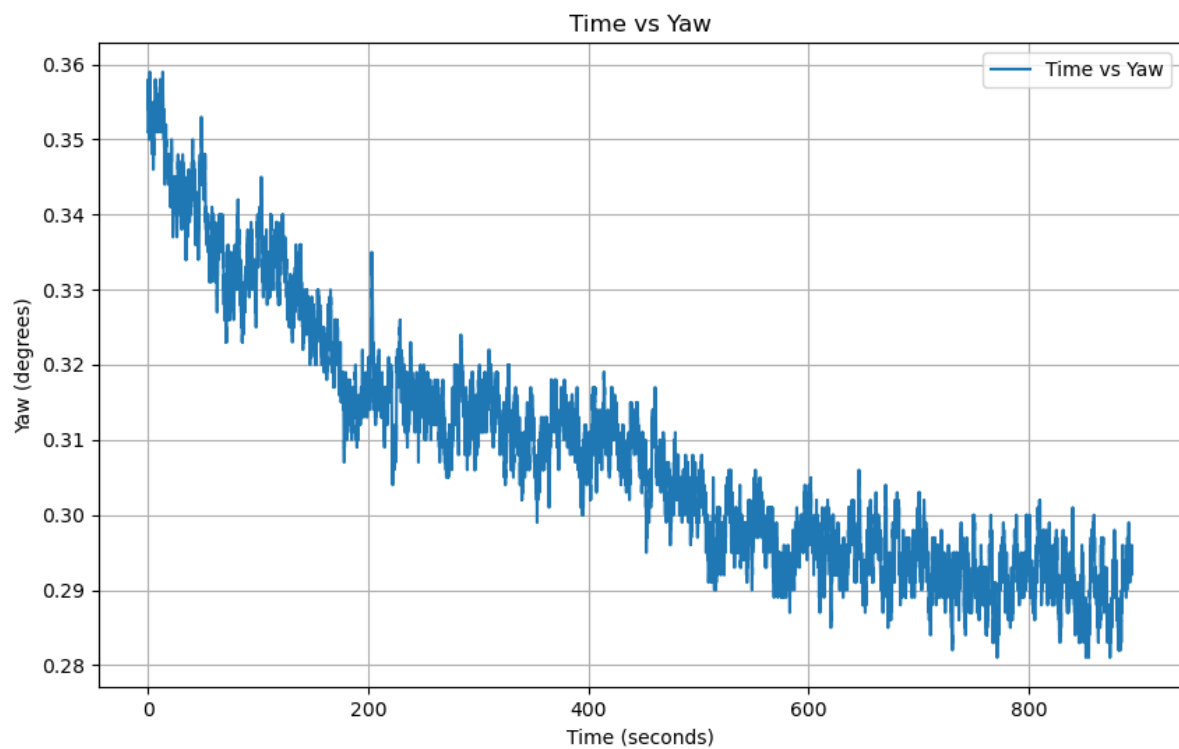


Figure 12: Time series data for Yaw.

- **Frequency Distribution Plots:**
 - **Linear Acceleration (X, Y, Z)**

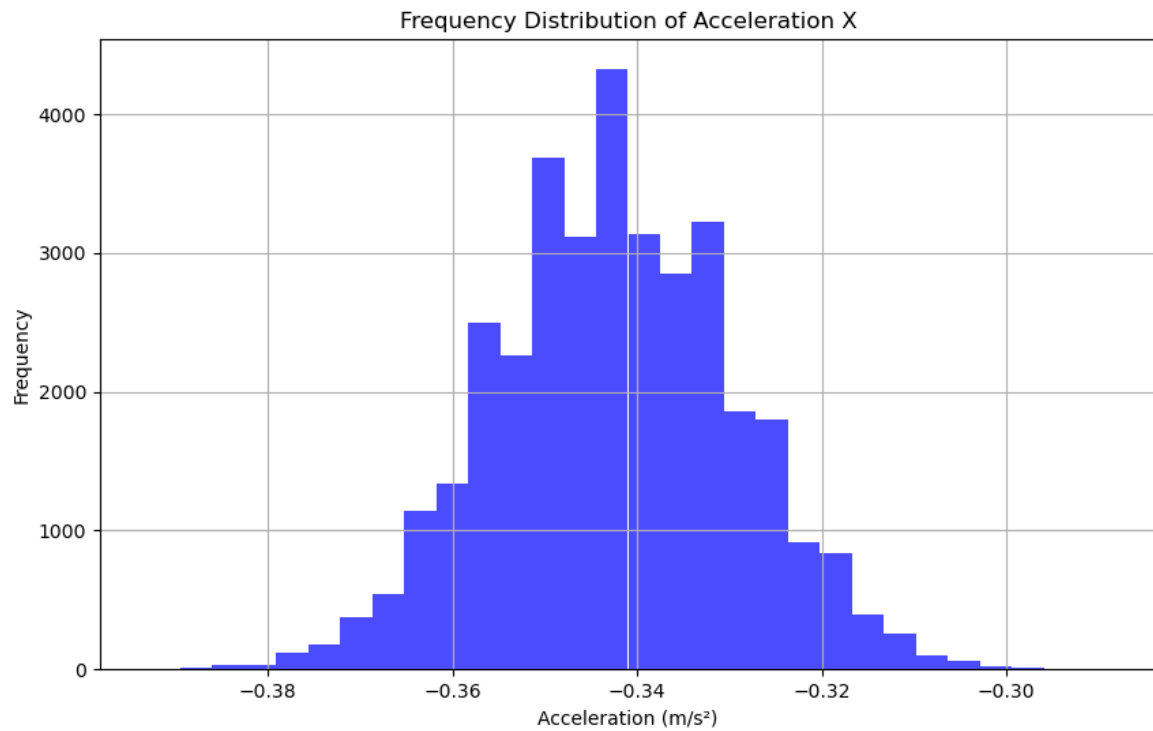


Figure 13: Frequency Distribution for accelerometer X-axis.

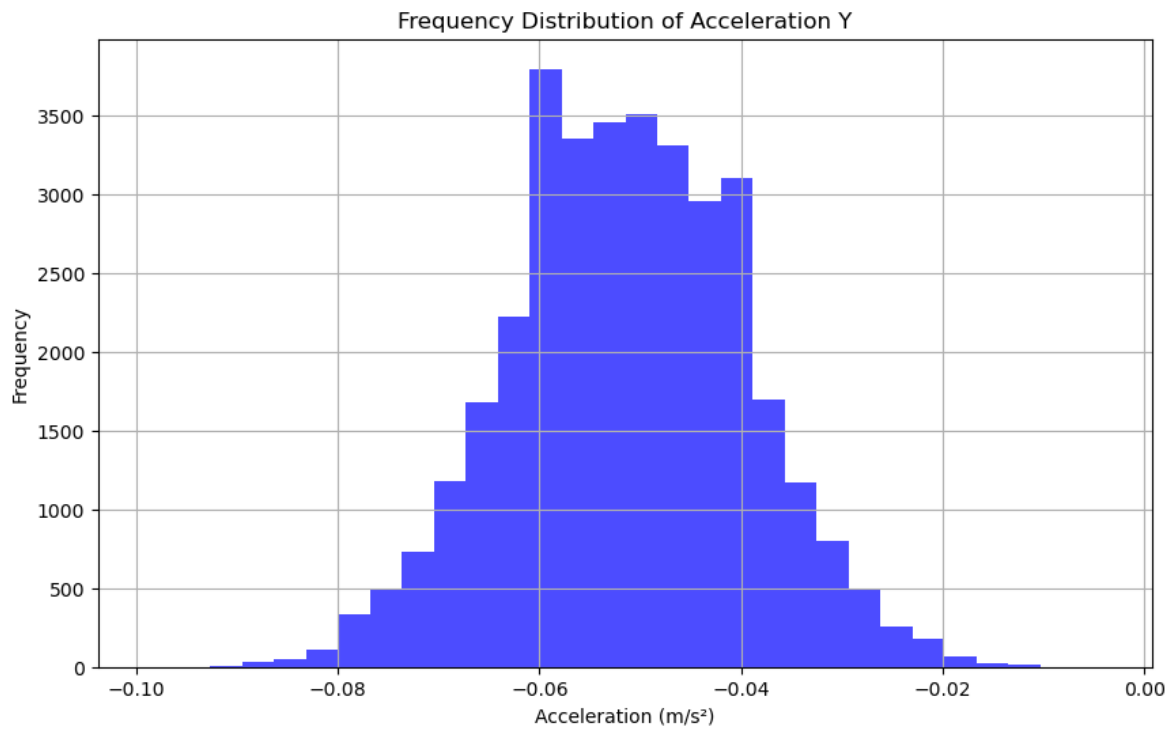


Figure 14: Frequency Distribution for accelerometer Y-axis.

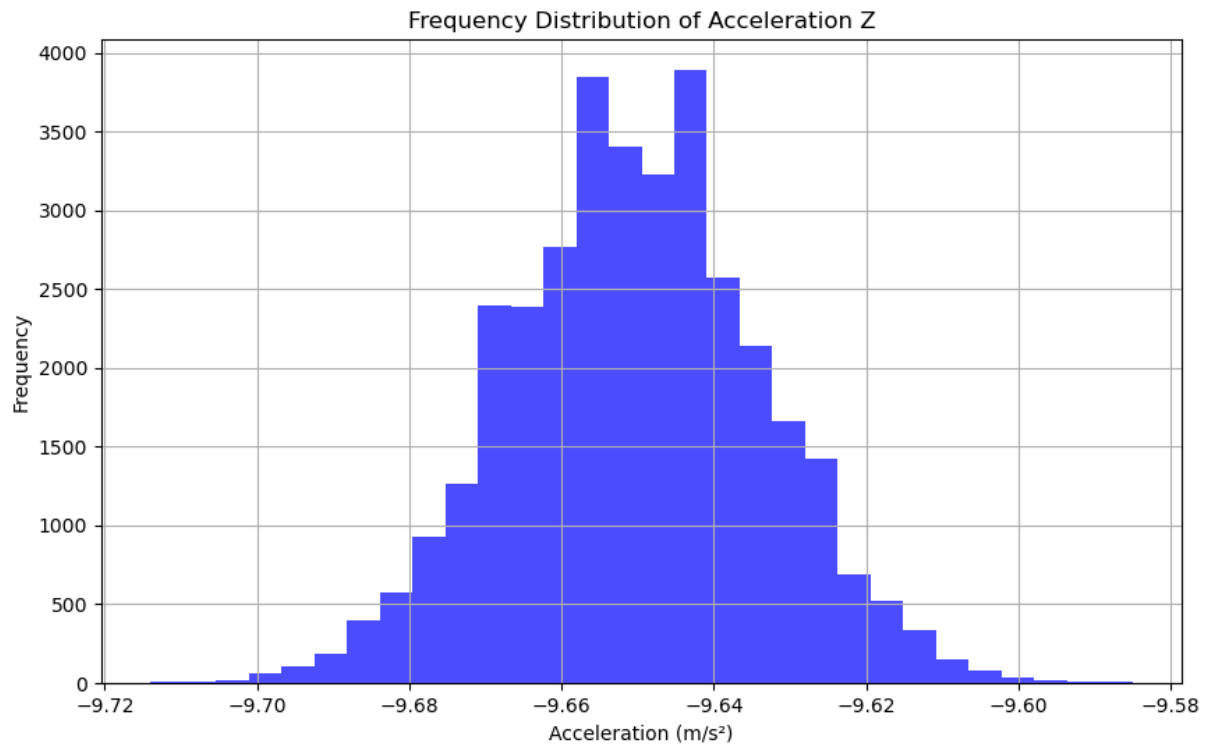


Figure 15: Frequency Distribution for accelerometer Z-axis.

- **Angular Velocity (X, Y, Z)**

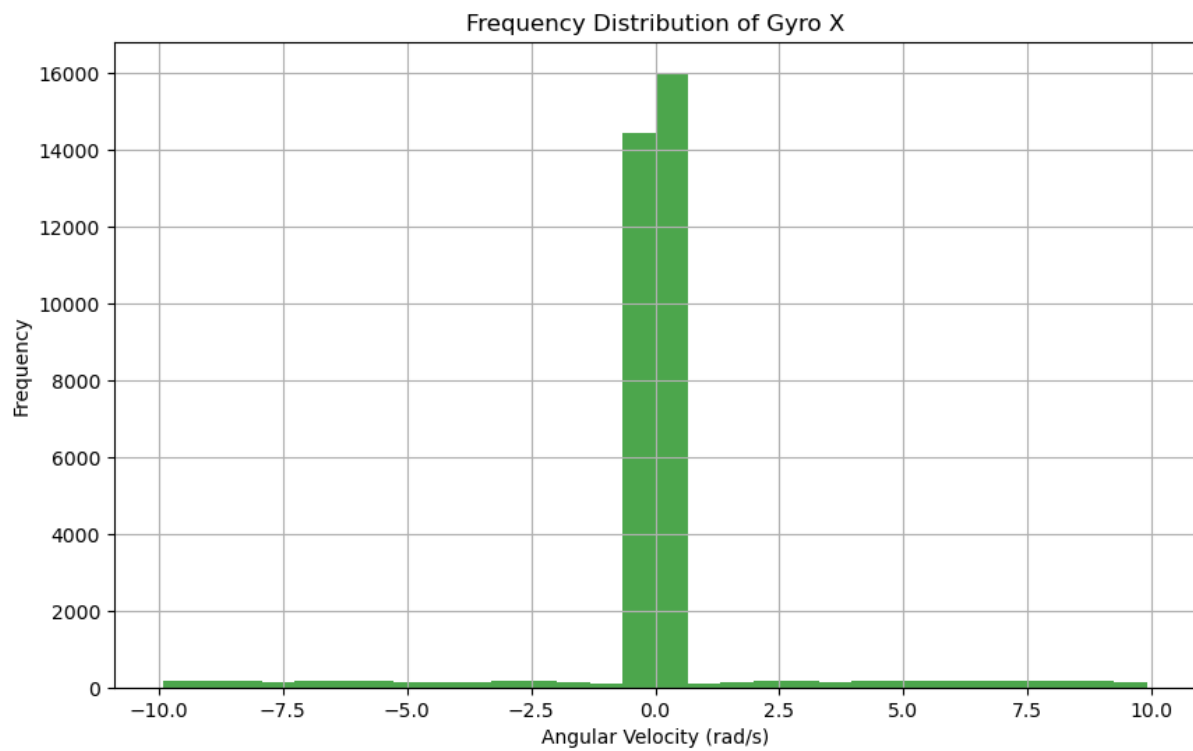


Figure 16: Frequency Distribution for Gyro X-axis.

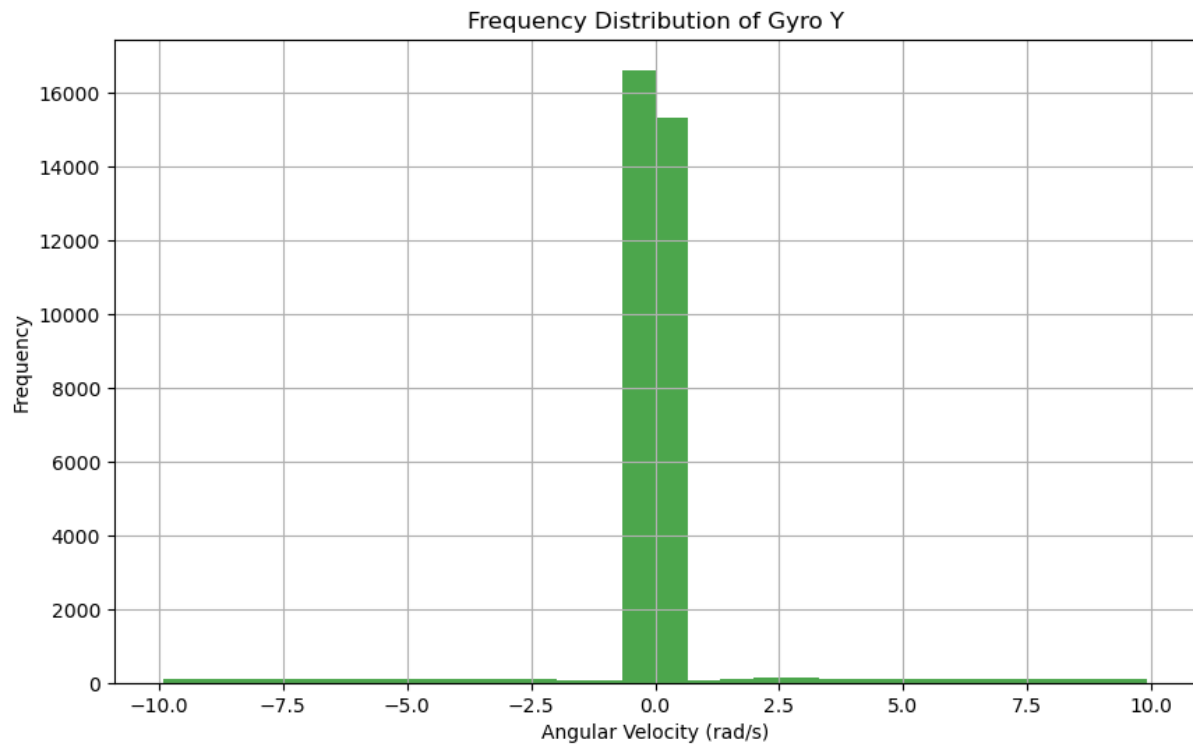


Figure 17: Frequency Distribution for Gyro Y-axis.

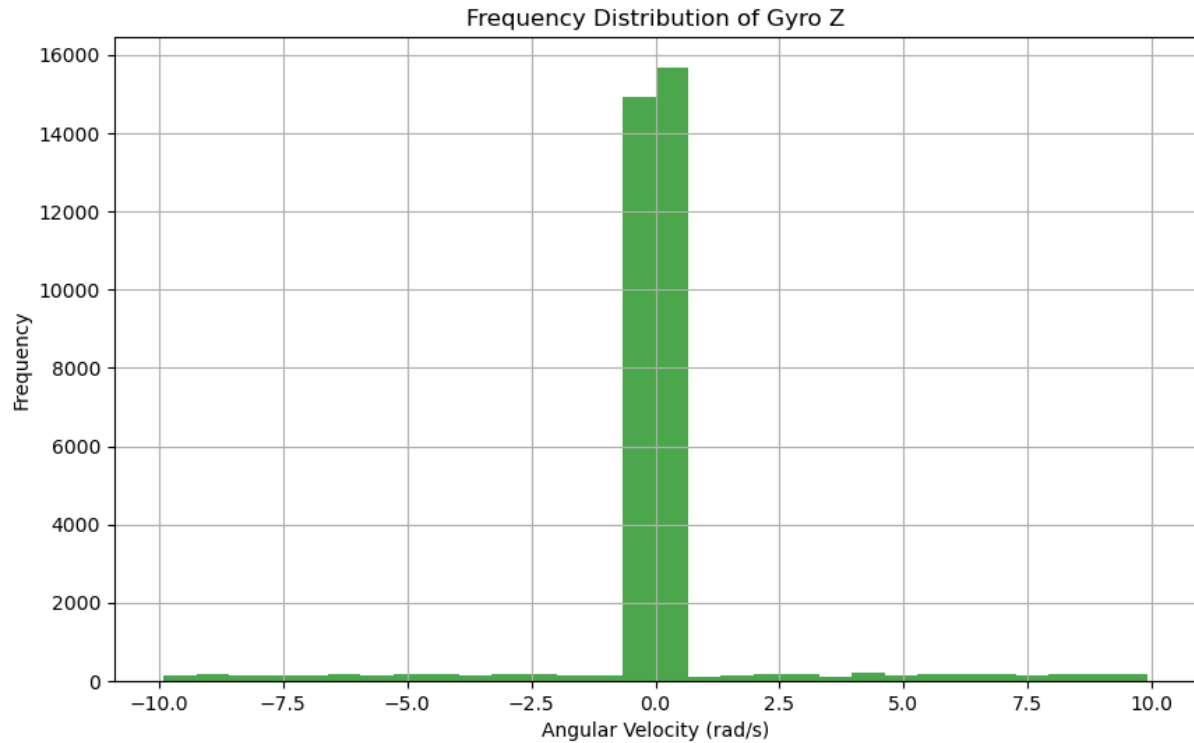


Figure 18: Frequency Distribution for Gyro Z-axis.

- **Magnetic Field (X, Y, Z)**

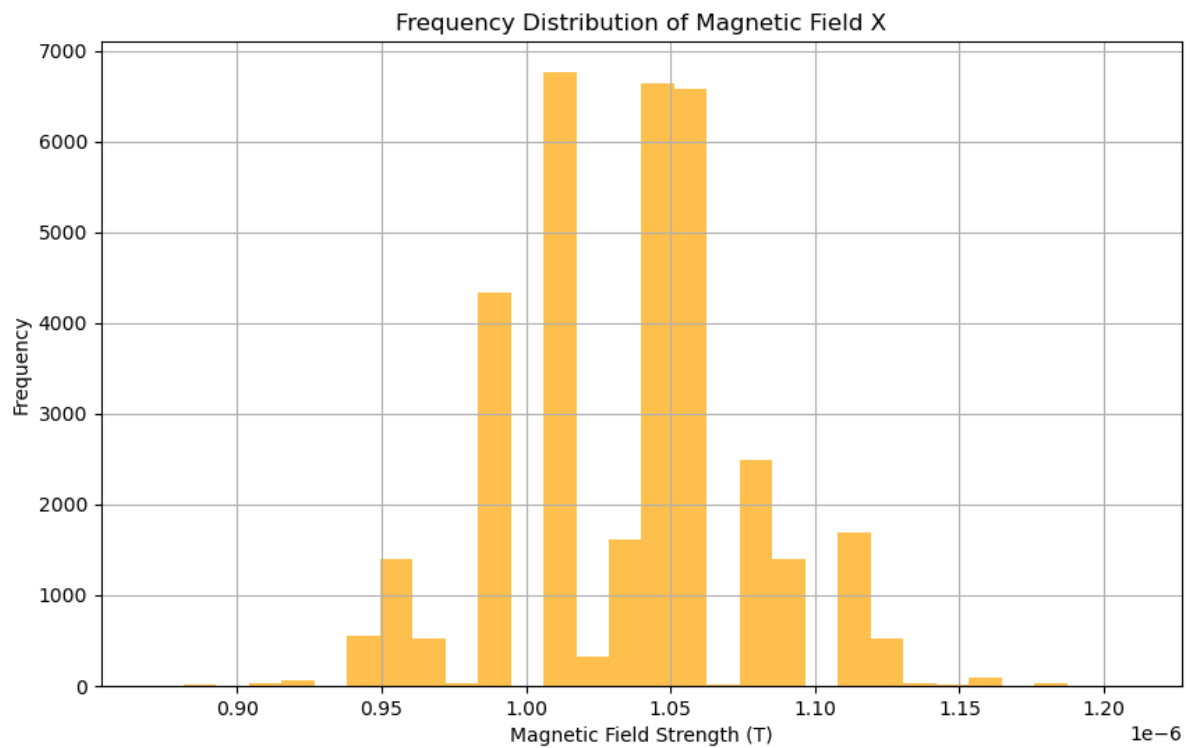


Figure 19: Frequency Distribution for Magnetic Field X-axis.

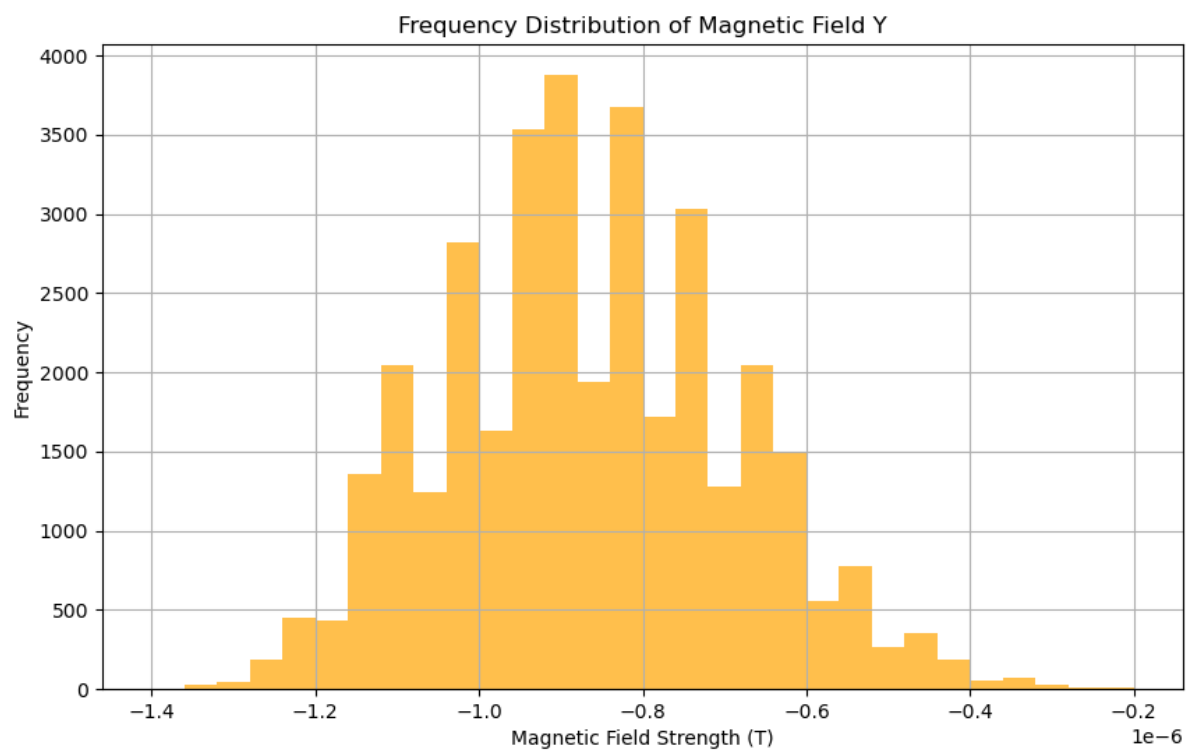


Figure 20: Frequency Distribution for Magnetic Field Y-axis.

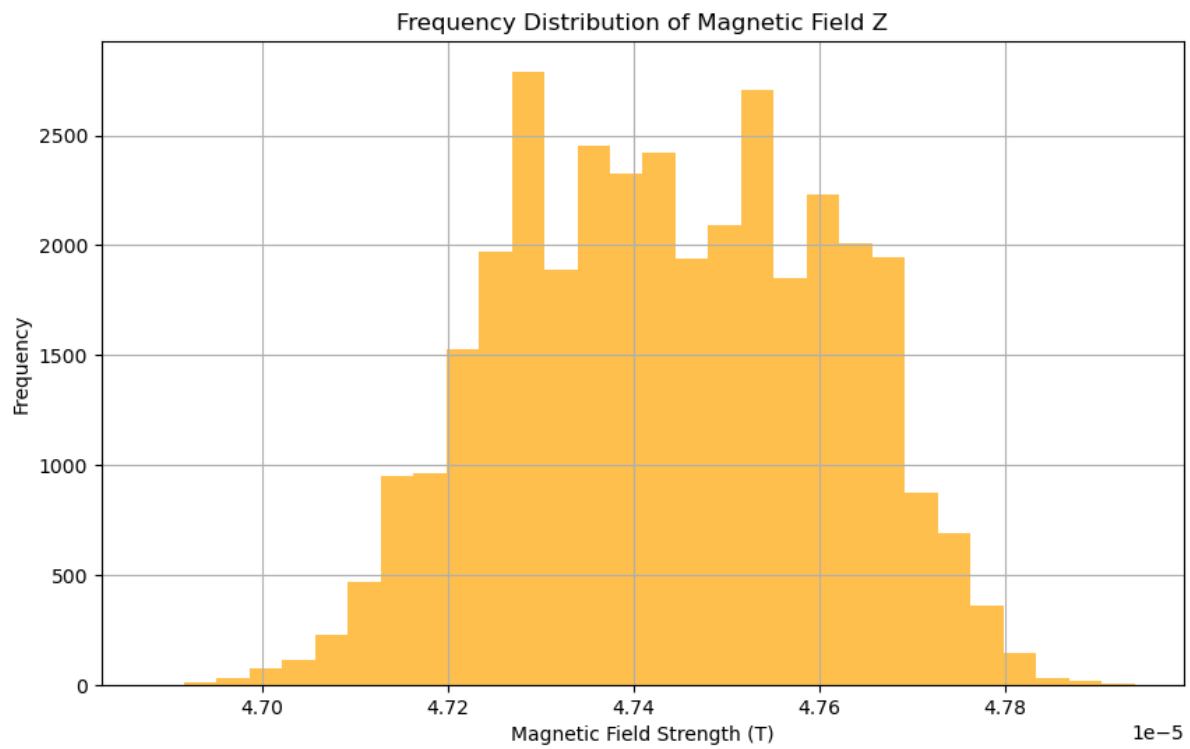


Figure 21: Frequency Distribution for Magnetic Field Z-axis.

- **Orientation (Roll, Pitch, Yaw)**

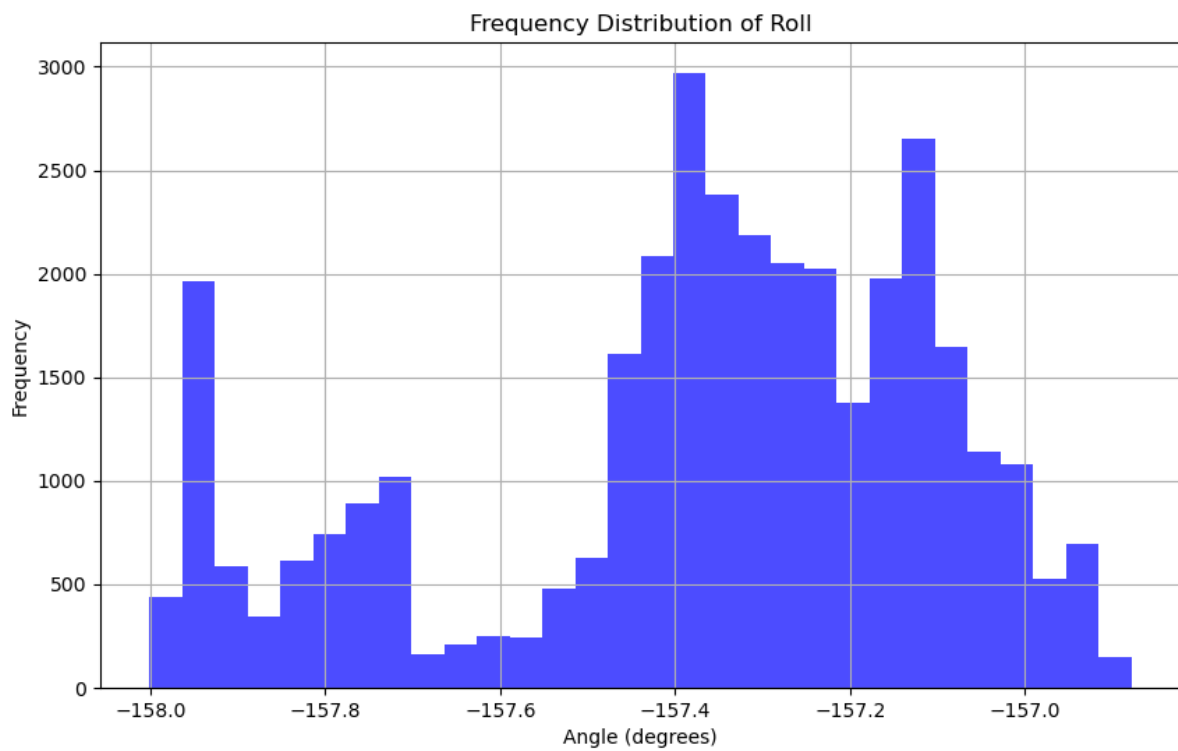


Figure 22: Frequency Distribution for Roll.

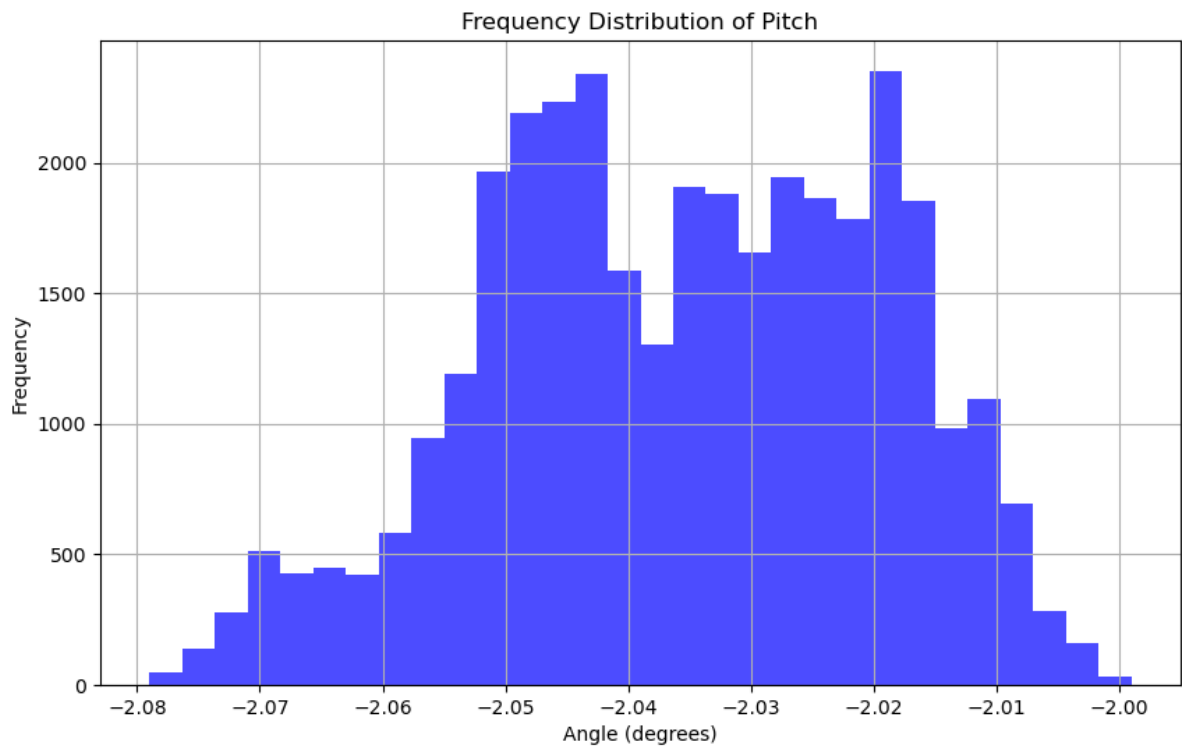


Figure 23: Frequency Distribution for Pitch.

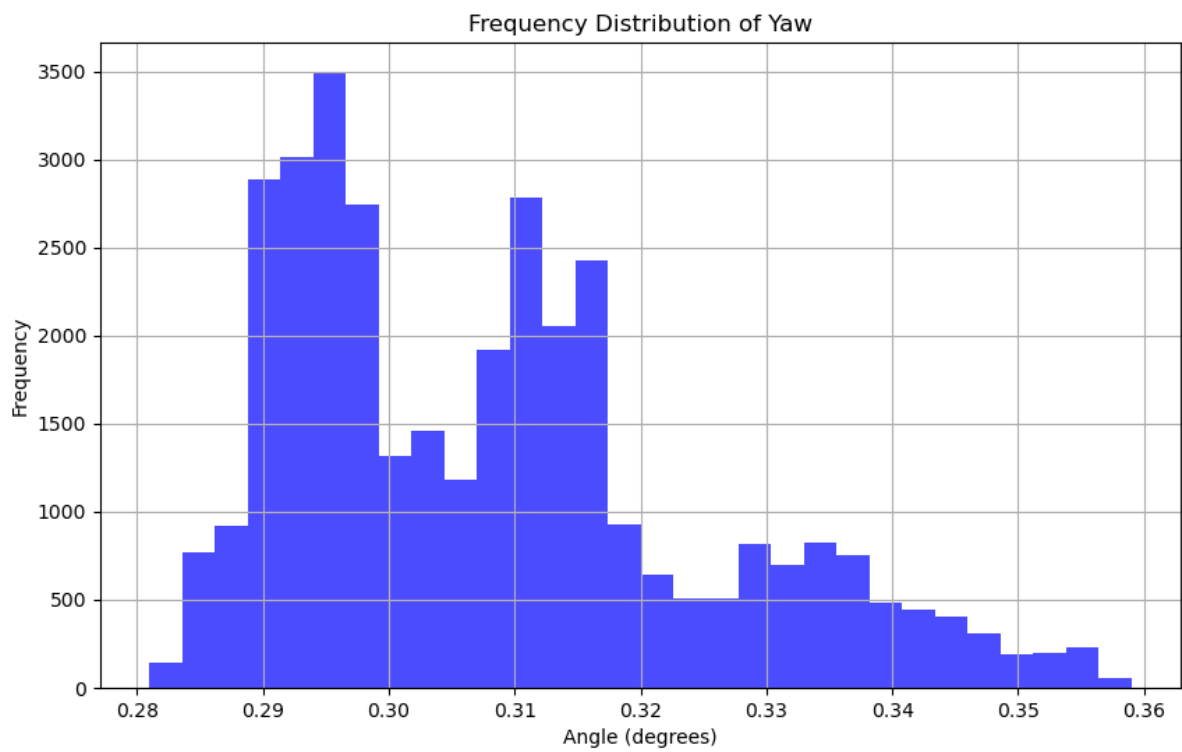


Figure 24: Frequency Distribution for Yaw.

Table:

	Mean	Standard Deviation
<i>Accelerometer</i>		
Linear Acceleration X	-0.342402	0.012994
Linear Acceleration Y	-0.051443	0.011708
Linear Acceleration Z	-9.650426	0.016608
<i>Gyroscope</i>		
Angular Velocity X	0.003045	2.187243
Angular Velocity Y	0.000366	1.787328
Angular Velocity Z	0.005834	2.124165
<i>Magnetometer</i>		
Magnetic Field X	0.000001	0
Magnetic Field Y	-0.000001	0
Magnetic Field Z	0.000047	0
<i>Orientations</i>		
Roll	-157.371352	0.276571
Pitch	- 2.035495	0.015968
Yaw	0.308244	0.016528

- **Allan Deviation Plots:**

- **Linear Acceleration (X, Y, Z)**

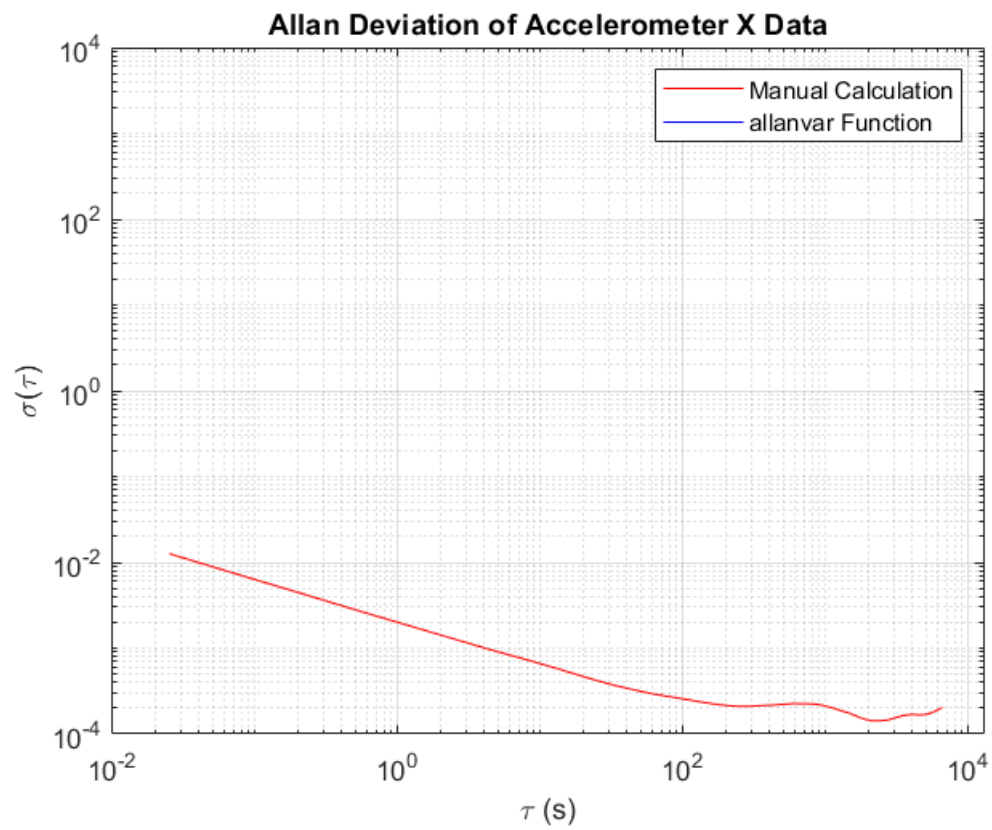


Figure 25: Allan Deviation Of Accelerometer along X.

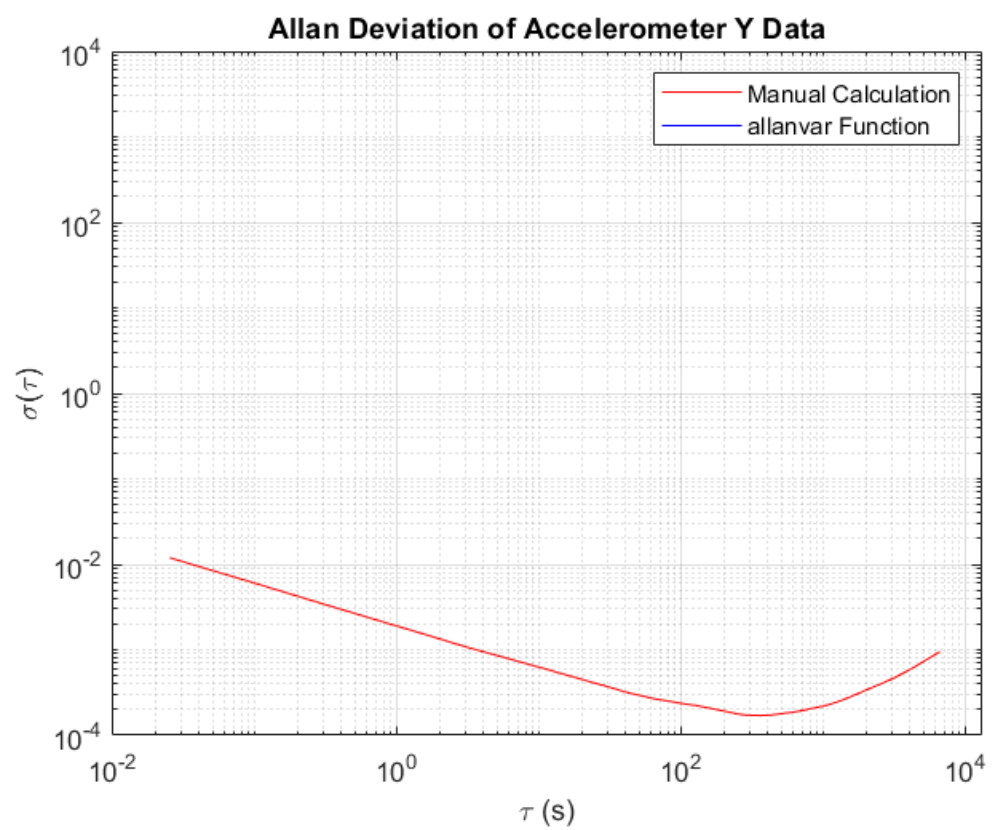


Figure 26: Allan Deviation Of Accelerometer along Y.

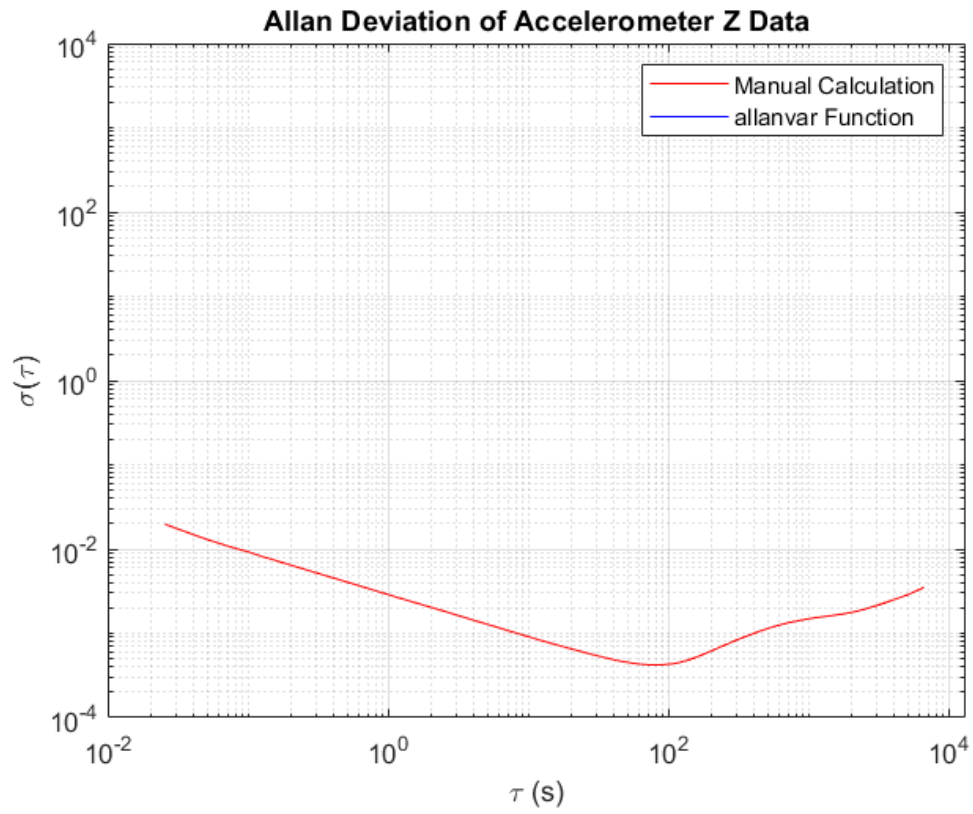


Figure 27: Allan Deviation Of Accelerometer along Z.

○ **Angular Velocity (X, Y, Z)**

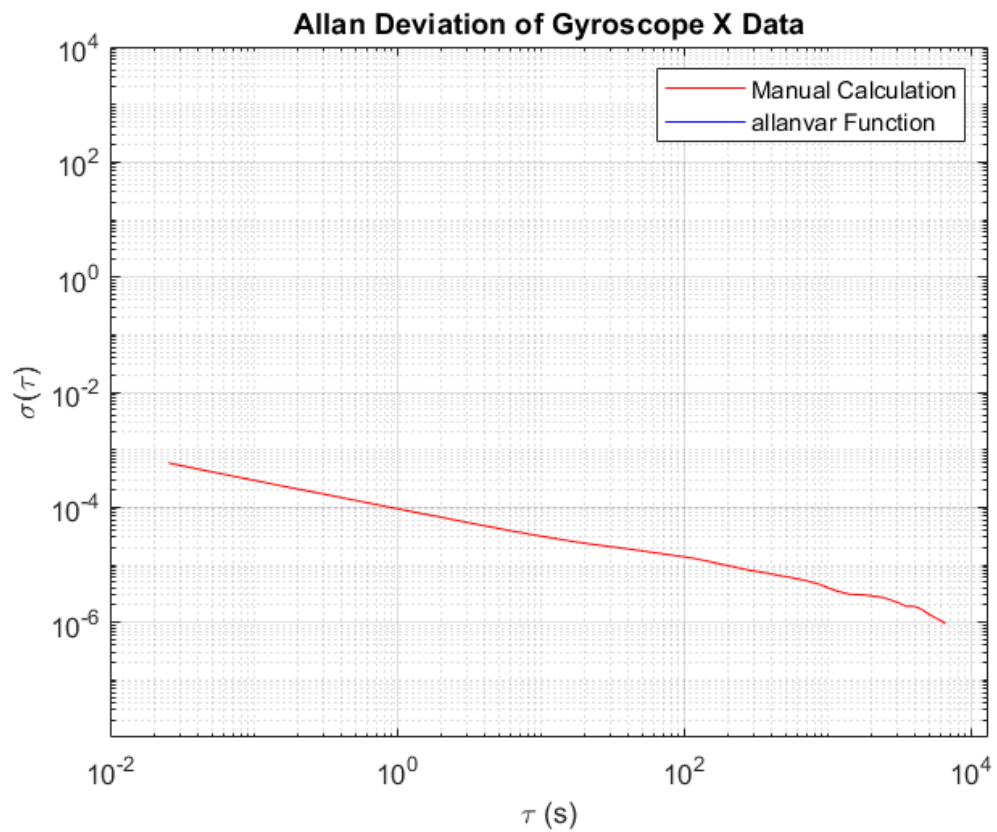


Figure 28: Allan Deviation Of Gyroscope along X.

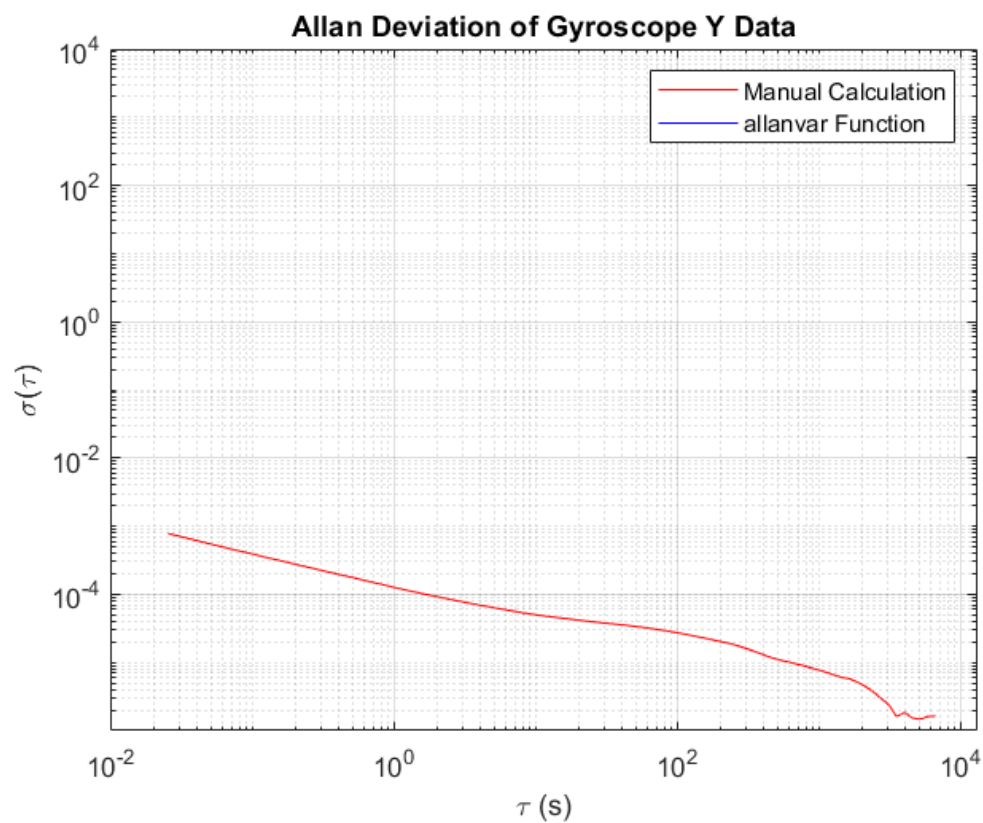


Figure 29: Allan Deviation Of Gyroscope along Y.

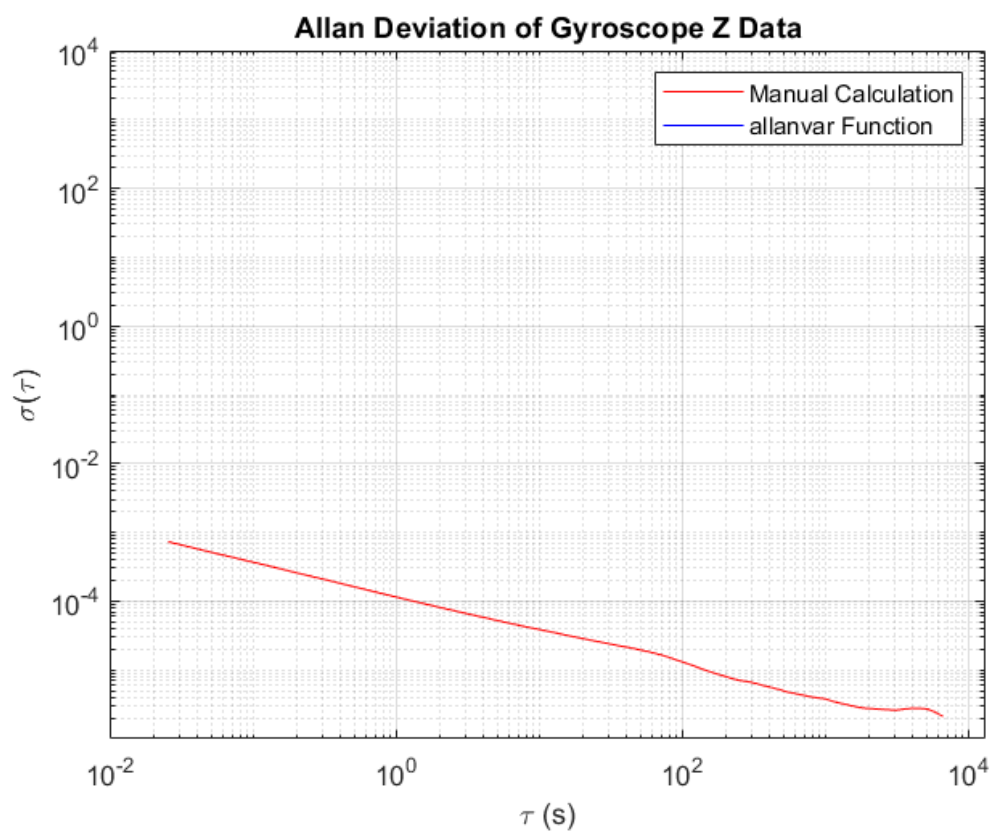


Figure 30: Allan Deviation Of Gyroscope along Z.

- **Noise Parameter Plots:**

- Linear Acceleration (X, Y, Z)

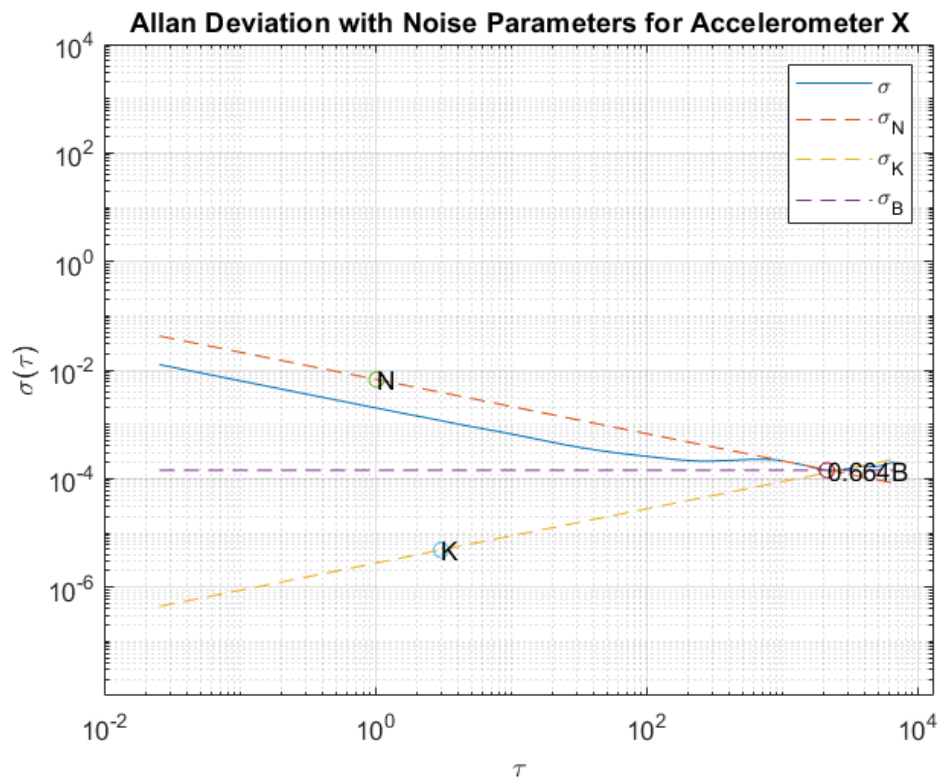


Figure 31: Allan Deviation with Noise Parameter of Accelerometer along X.

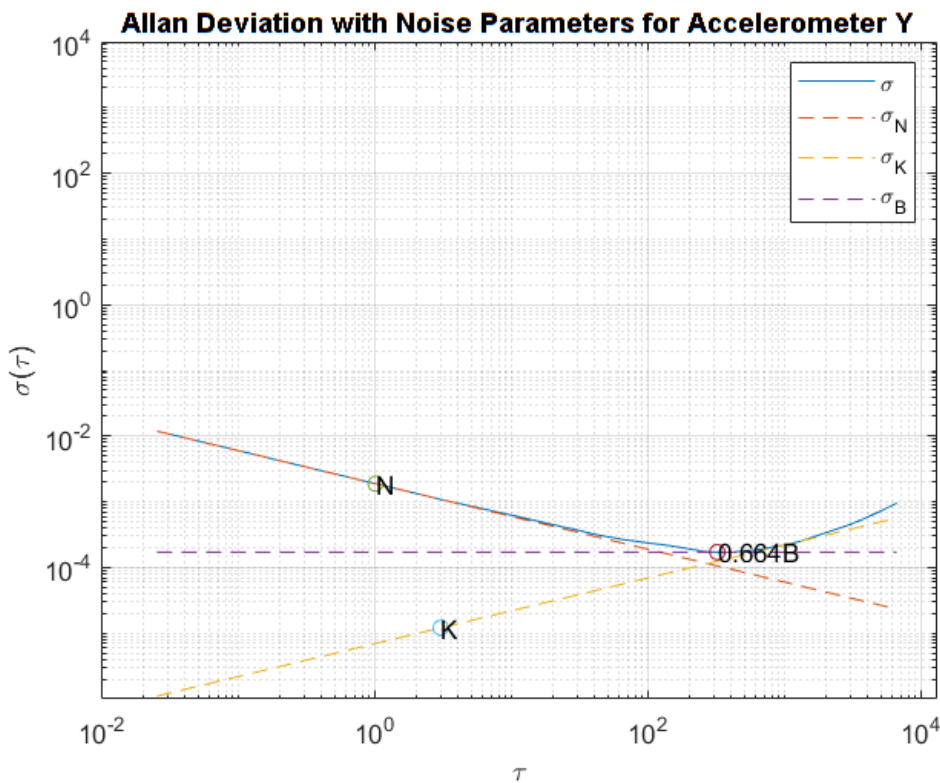


Figure 32: Allan Deviation with Noise Parameter of Accelerometer along Y.

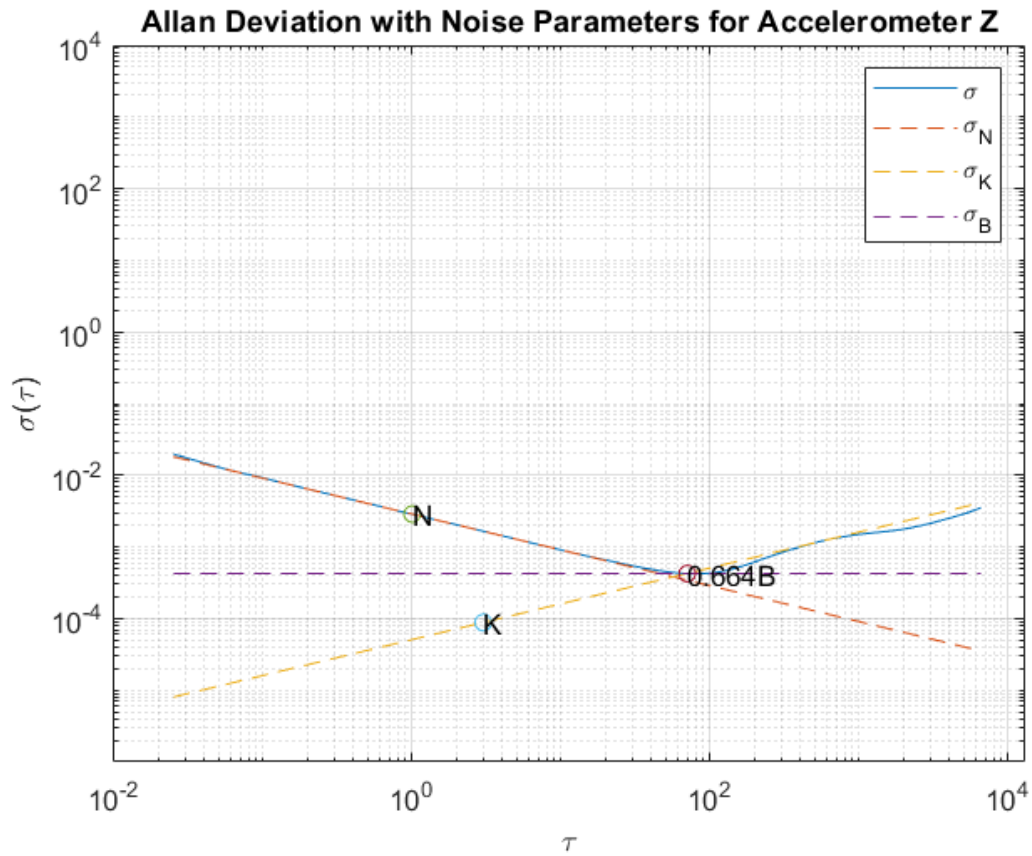


Figure 33: Allan Deviation with Noise Parameter of Accelerometer along Z.

- Angular Velocity (X, Y, Z)

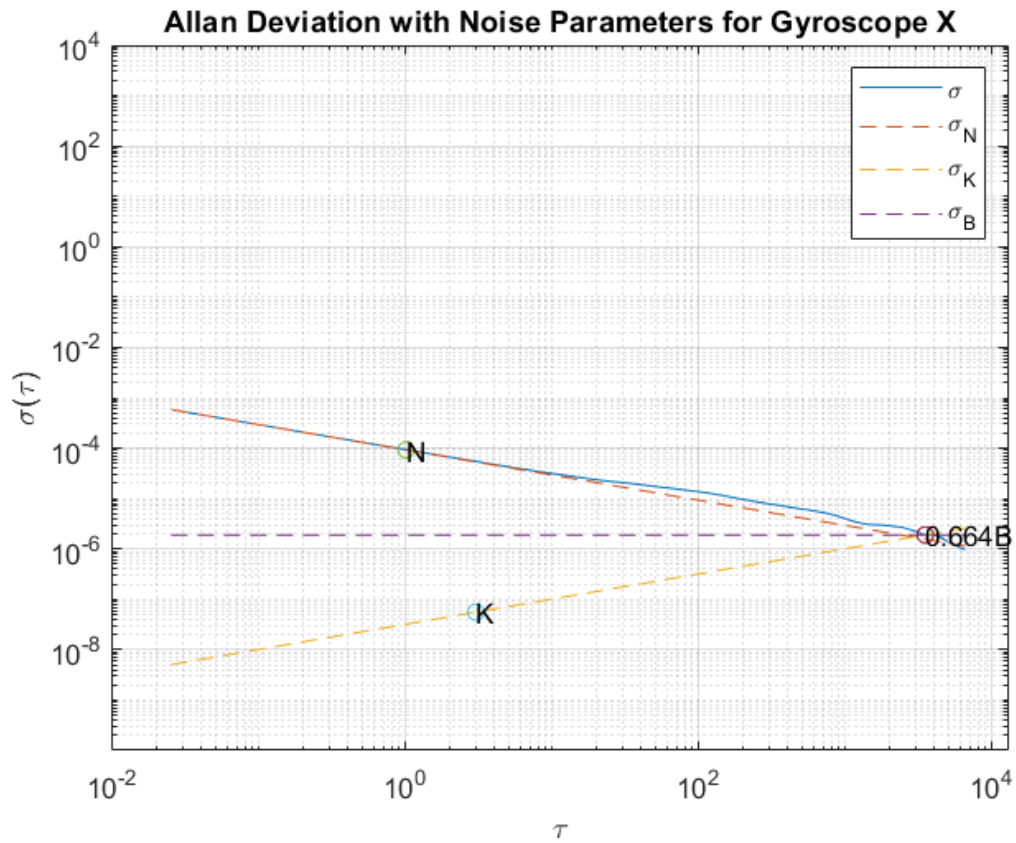


Figure 34: Allan Deviation with Noise Parameter of Gyroscope along X.

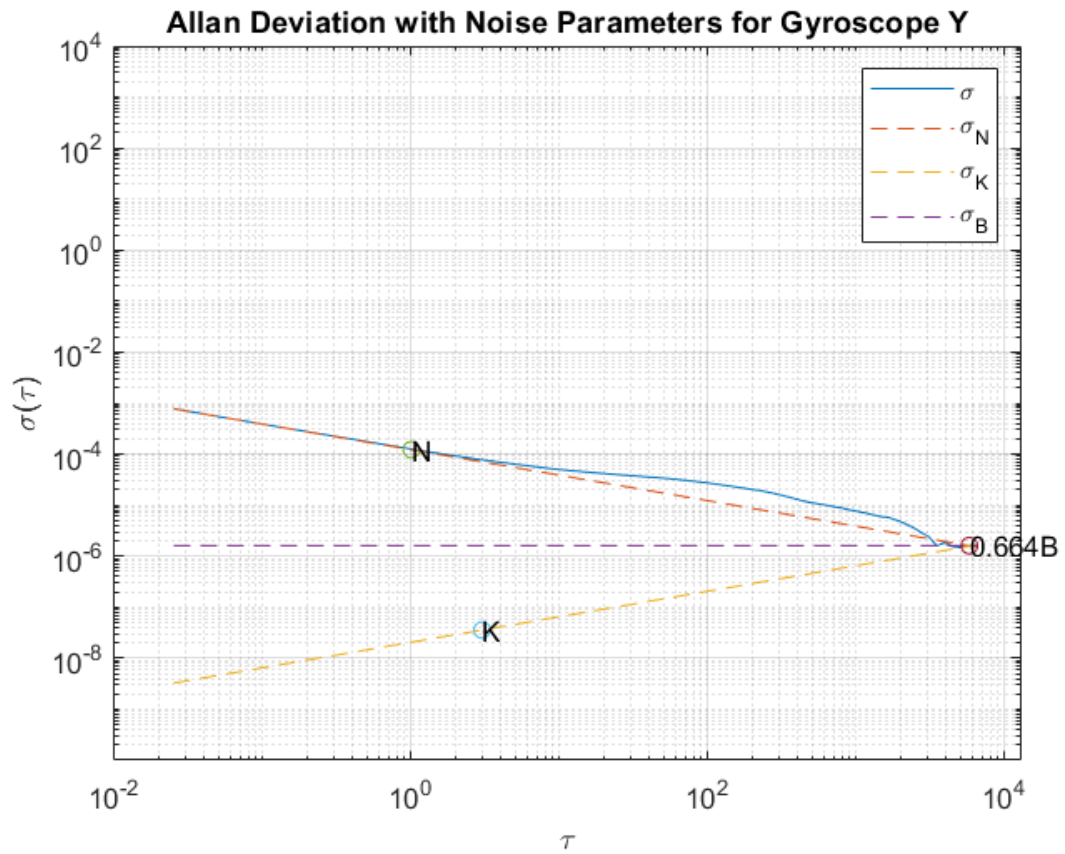


Figure 35: Allan Deviation with Noise Parameter of Gyroscope along Y.

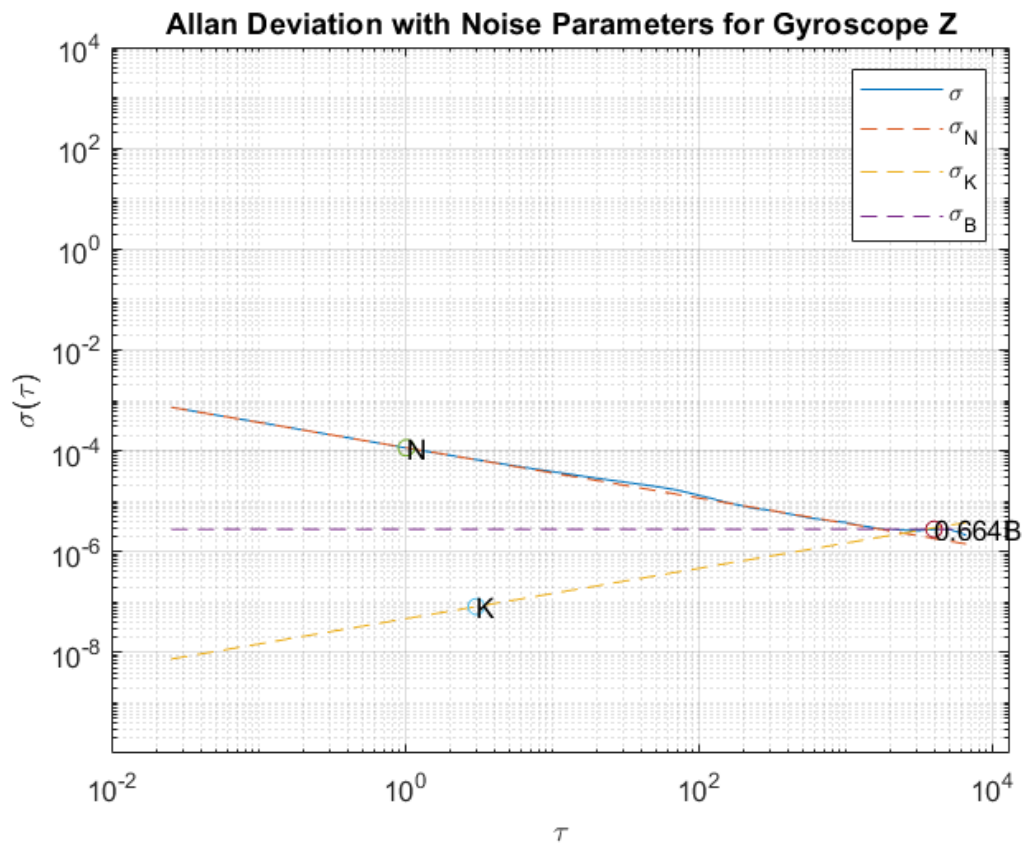


Figure 36: Allan Deviation with Noise Parameter of Gyroscope along Z.

Table:

	N	K	B
Gyroscope			
X	9.23596 e-05	5.55549 e-08	2.85246 e-06
Y	0.000122056	3.56721 e-08	2.42566 e-06
Z	0.000114535	8.1196 e-08	4.1708 e-06
Accelerometer			
X	0.0066851	4.8395 e-06	2.14193 e-04
Y	0.0018867	1.20967 e-05	2.57955 e-04
Z	0.00286417	8.81869 e-05	6.37883 e-04

Conclusions :

Figure 1 to Figure 3: This figure shows the time series plot of the linear acceleration along the axes of the accelerometer. The plot is centered around 0 m/s², indicating that the accelerometer is stationary. The data exhibits random fluctuations, suggesting the presence of noise in the measurements.

Figure 4 to Figure 6: This plot shows the angular velocity along the X-axis, Y-axis and Z-axis of the gyroscope, centered around 0 rad/s, indicating a stationary state with noise.

Figure 7 to Figure 9: These plots show the magnetic field data, centered around a non-zero value, indicating a constant magnetic field affecting the sensor.

Figure 10 to Figure 12: These plots show stable data with small fluctuations, indicating a stationary sensor.

Figure 13 to Figure 24: The distributions are characterized by a bell-shaped curve. The mean of each distribution represents the central tendency of the data. The standard deviation (or variance) represents the spread or dispersion of the data around the mean. The frequency distribution plots also suggest the presence of white noise.

The presence of both Gaussian and white noise suggests that the sensor's errors have both random (Gaussian) and frequency-independent (white) components. This information is crucial for developing appropriate filtering and compensation techniques to improve the sensor's performance and reliability in robotics applications. The consistency of these noise characteristics across different measurement types (acceleration, angular velocity, magnetic field, and orientation) indicates a uniform quality across all sensor components.

Figure 25 to Figure 36: The plots show a "V-shape", indicating white noise and random walk noise processes. The Allan Deviation plots for the accelerometer data (X, Y, and Z axes) and gyroscope data (X, Y, and Z axes) exhibit a characteristic "V-shape". This shape is indicative of two primary noise processes:

1. White Noise: Dominant at shorter time scales, represented by the descending slope on the left side of the "V".
2. Random Walk (or Brownian Motion) Noise: Dominant at longer time scales, represented by the ascending slope on the right side of the "V".

Significance of the Results:

While specific numerical values for these parameters were not provided in the summary, the analysis reveals several important points:

1. **Comprehensive Noise Characterization:** The presence of both white noise and random walk processes in both the accelerometer and gyroscope data provides a comprehensive picture of the sensor's noise characteristics.
2. **Performance Across Time Scales:** The "V-shape" of the Allan Deviation plots indicates that the sensor performs differently at short and long time scales, which is crucial information for applications requiring either quick measurements or long-term stability.
3. **Quantitative Performance Metrics:** The estimation of VRW/ARW and Bias Instability provides quantitative metrics that can be used to compare this sensor's performance with other IMUs or to determine its suitability for specific applications.
4. **Sensor Quality:** The fact that these noise processes are well-characterized and follow expected patterns suggests that the VectorNav VN-100 IMU is a high-quality sensor with predictable and consistent performance.

In conclusion, the Allan Variance analysis and noise parameter estimation provide a detailed and quantitative understanding of the VectorNav VN-100 IMU's performance characteristics. This information is invaluable for determining the sensor's suitability for various applications and for implementing appropriate filtering and estimation algorithms to optimize its performance in real-world scenarios.

Answering the Initial Questions :

1. What Kind of Errors/Sources of Noise Are Present?

-> In inertial sensors like gyroscopes and accelerometers, several types of errors and noise sources can be present. These include:

Gyroscope Errors/Noise:

Angle Random Walk (ARW): This is white noise in the gyroscope measurements, leading to random fluctuations.

Rate Random Walk (RRW): This is caused by random drift in the gyroscope's bias over time.

Bias Instability: This is a low-frequency noise causing slow variations in the gyroscope's bias.

Quantization Noise: This arises from the discrete nature of digital sensors.

Temperature Drift: Variations in temperature can cause changes in the sensor's bias and scale factor.

Vibration-Induced Errors: High-frequency vibrations can introduce errors in the gyroscope measurements.

Accelerometer Errors/Noise:

Velocity Random Walk (VRW): This is the accelerometer's equivalent of ARW, representing white noise in the acceleration measurements.

Bias Instability: Similar to gyroscopes, this causes slow variations in the accelerometer's bias.

Scale Factor Instability: Changes in the sensor's sensitivity over time or temperature variations.

Misalignment Errors: Due to imperfect alignment of the accelerometer axes.

Temperature Effects: Variations in temperature can affect the accelerometer's bias and scale factor.

2. How Do We Model Them? Where Do We Measure Them?

-> To model these errors, we use various noise parameters derived from Allan variance analysis:

Angle Random Walk (N): Characterized by the slope of $-0.5-0.5-0.5$ in the Allan deviation plot for gyroscopes.

Rate Random Walk (K): Characterized by the slope of $0.50.50.5$ in the Allan deviation plot for gyroscopes.

Bias Instability (B): Characterized by the flat region (slope of 000) in the Allan deviation plot.

Velocity Random Walk (VRW): Similar to ARW but for accelerometers.

Bias Instability for Accelerometers: Similar to gyroscopes.

Measurement Process:

Data Collection: Log stationary data from the gyroscope and accelerometer for a significant duration (e.g., 5 hours).

Allan Variance Analysis: Perform Allan variance analysis on the logged data to obtain the Allan deviation plots.

Extract Noise Parameters:

Identify segments of the Allan deviation plot with slopes $(-0.5 -0.5 -0.5)$, $(0.5 0.5 0.5)$, and $(0 0 0)$ to estimate ARW, RRW, and bias instability respectively.

Calculate the corresponding noise parameters NNN, KKK, and BBB.